

Case Document 1: Geriatric Acute Respiratory Failure with Cardiac Overload

Patient Identification: Mr. Arthur Penhaligon, 82-year-old Male.

Admission Date: 10 December 2024.

History of Present Illness: The patient presented to the emergency department with a four-day history of worsening dyspnea, orthopnea, and abdominal distension accompanied by significant bilateral lower limb swelling.¹⁴ He has an extensive past medical history including Type 2 Diabetes Mellitus, Ischaemic Heart Disease (IHD), Chronic Obstructive Lung disease (COL/COPD), and chronic hypertension.⁶

Hospital Course: On arrival, the patient was found to be in respiratory distress. Vital signs revealed a BP of 165/95 mmHg and an oxygen saturation of 88% on room air. Chest X-ray demonstrated bilateral pleural effusions and cardiomegaly. He was managed with aggressive diuresis and empirical antibiotic coverage for a suspected secondary chest infection complicating his underlying COL.

Discharge Medications:

- **Ofloxacin:** 0.4g administered Intravenously (IV) twice daily (BD) for 3 days.
- **Cefepime:** 1g administered Intravenously (IV) twice daily (BD) for 5 days.
- **Azithromycin:** 0.5g administered Orally (Oral) once daily (OD) for 3 days.
- **Indication:** Treatment for abdominal distension and suspected chest infection in a patient with chronic cardiac and pulmonary failure.⁶

Case Document 2: Acute Dysentery and Septic Shock in a Young Adult

Patient Identification: Ms. Sarah Thompson, 22-year-old Female.

Admission Date: 15 January 2025.

History of Present Illness: A 22-year-old female was admitted in a state of septic shock following a 48-hour history of high-grade fever, abdominal cramping, and tenesmus with bloody, mucoid diarrhea.⁶ Initial assessment in the triage bay showed a heart rate of 125 bpm and a temperature of 39.5°C.

Clinical Management: Aggressive fluid resuscitation was initiated. Stool cultures were obtained, and the patient was started on dual IV therapy for suspected acute bacillary dysentery. By hospital day 3, the patient showed signs of clinical improvement, with cessation of fever and stabilization of blood pressure.

Discharge Medications (Transitioning to Outpatient):

- **Cefoperazone+Sulbactam:** 1g IV twice daily (BD) for 5 days.
- **Metronidazole:** 0.5g IV three times daily (TDS) for 5 days.
- **Cefixime:** 0.2g Orally (Oral) twice daily (BD) for 3 days post-discharge.
- **Metronidazole:** 0.5g Orally (Oral) three times daily (TDS) for 3 days.⁶

Case Document 3: Post-Operative Management in a Frail Diabetic Patient

Patient Identification: Mrs. Gladys Henderson, 87-year-old Female.

Admission Date: 22 February 2025.

History of Present Illness: The patient underwent an emergency below-knee (BE) amputation due to advanced Peripheral Artery Disease (PAD) of the left lower limb that resulted in non-healing ulcers and gangrene.⁶

Hospital Course: Post-operative recovery was complicated by the patient's poorly controlled Type 2 Diabetes and a pre-existing history of tuberculosis, for which she was currently receiving AKT (Anti-Koch's Treatment).⁶ Due to the high risk of surgical site infection, she received a short course of prophylactic coverage alongside her ongoing anti-tubercular regimen.

Medication Summary:

- **Ceftriaxone:** 1g IV twice daily (BD) for 1 day.
- **Ongoing Medications:** Continuation of the standard AKT-4 Kit (Rifampicin, Isoniazid, Ethambutol, Pyrazinamide).²⁴
- **Indication:** Post-operative prophylaxis and continuation of therapy for Koch's lung.⁶

Case Document 4: Acute Aspiration Pneumonia Following Cerebral Infarct

Patient Identification: Mr. Samuel Wick, 65-year-old Male.

Admission Date: 05 March 2025.

History of Present Illness: The patient was brought to the hospital after being found unconscious at home. Neuroimaging confirmed an acute cerebral infarct in the right middle cerebral artery territory.⁶ He has a history of an "old stroke" and subsequent neurological deficits.

Hospital Course: During the acute phase of his stroke, the patient experienced an episode of vomiting and decreased consciousness, leading to clinical concerns regarding aspiration. Physical exam revealed coarse crackles in the right lung base.

Antibiotic Regimen:

- **Ceftriaxone:** 1g IV twice daily (BD) for 1 day.
- **Metronidazole:** 0.2g Orally (Oral) three times daily (TDS) for 5 days.
- **Cefixime:** 0.02g Orally (Oral) twice daily (BD) for 5 days.
- **Note:** The intravenous therapy was quickly transitioned to oral as the patient's gag reflex improved and swallowing assessments were completed.⁶

Case Document 5: Retroviral Infection with Opportunistic Lung Pathology

Patient Identification: Ms. Elena Vance, 31-year-old Female.

Admission Date: 12 April 2025.

History of Present Illness: The patient is a 31-year-old female known to be living with RVI (Retroviral Infection) Stage 4.⁶ She presented with a two-week history of non-productive cough, night sweats, and "art noise" (Antiretroviral Therapy related side effects).

Clinical Assessment: Labs indicated significant bone marrow depression with neutropenia. Imaging showed bilateral interstitial infiltrates, suspicious for *Pneumocystis jirovecii* pneumonia (PCP).⁶ There was also evidence of underlying hepatitis likely secondary to chronic viral coinfection.

Therapeutic Intervention:

- **Septrin (Co-trimoxazole):** 0.96g Orally (Oral) once daily (OD) for 1 day.
- **Ceftriaxone:** 1g IV twice daily (BD) for 5 days.
- **Cefepime:** 1g IV twice daily (BD) for 4 days.

Indication: Management of opportunistic infection risk and clinical stabilization in the context of advanced RVI and bone marrow suppression.